Paper Review • Physical AI / Continual Learning

DRAP: Plasticity Scaled by Physical Risk, Less VLA Forgetting Without Replay

DRAP: Dynamics-Risk Adaptive Plasticity for Continual Learning in Physical AI

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Code: github.com/oswaldoludwig/DRAP-Dynamics-Risk-Adaptive-Plasticity

In short: physical-risk importance cuts key tensors' LR

DRAP fits local linear dynamics to expert trajectories, uses **σ_max(A_t)** to gauge how much an error now would be physically amplified, weights per-sample action gradients by it into tensor-level importance, then directly lowers the **AdamW learning rate** of important tensors—no replay of old data, no stored reference weights.

0.77 → 0.31

Avg. forgetting after Goal

Std. FT 0.77 • KappaTune+EWC 0.63 • DRAP 0.31

0.55

Avg. success after Goal

Std. FT 0.26 • KappaTune+EWC 0.32

0.02

Spatial forgetting after Object

Std. FT 0.66 • KappaTune+EWC 0.35

Problem: in physical AI, small action errors snowball along trajectories

One action step
Small error
→
Next physical state drifts from demo
→
Observations fall out of distribution
→
Later errors amplify → Closed-loop failure

Sequential fine-tuning overwrites old skills; existing CL methods each have costs:

Experience replay

Strong, but must store or regenerate old observations

Expansion / routing (CLARE, Stellar VLA)

Isolates tasks, but adds modules or routing

Param. regularization (EWC, SI, MAS)

Protects old solutions, may limit new learning

Update suppression (CLASSP)

Lowers plasticity by past updates; may be too conservative for large models

DRAP's premise: parameters that support correct actions at dynamics-sensitive trajectory moments should become less plastic in later tasks.

Overview: KappaTune decides where to update, DRAP how much

Where

① KappaTune picks

- Score by condition number $\kappa(W)=\sigma max/(\sigma min+\varepsilon)$
- Fixed budget of low-κ tensors per vision/language/action head
- Selected once; rest frozen

How much

② Dynamics risk r_t

- Sliding-window ridge regression on expert trajectories fits A_t, B_t
- $\rho_t = \sigma_{max}(A_t)$
- Median-normalized, clipped to [0.25, 4]

How much

③ Importance I_T

- After task k, per-sample (batch=1) action-token gradients
- r_i^2-weighted mean squared gradient
- Normalized, added to history memory

How much

④ LR modulation

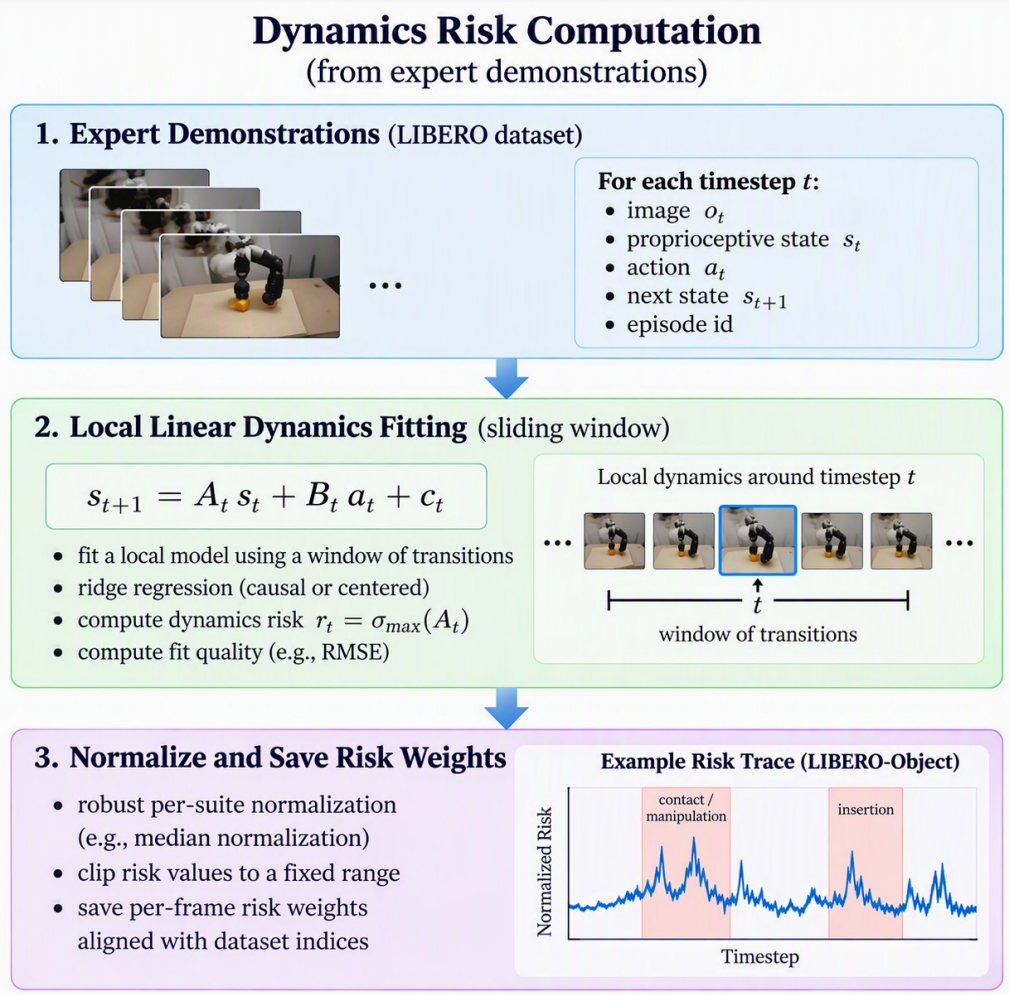
- $s_T = (1+\beta \bar{I}_T)^{(-1/p)}$
- $\eta_T = \eta$ • s_T (AdamW groups)
- Important tensors take small steps from a new task's first step

✓ No replay of old demos

✓ No stored reference weights θ*

✓ One importance scalar per tensor

Risk: local-linear σ_max(A_t) measures one-step amplification



1 Sliding-window local linear fit (per episode)

$\tilde{s}_{j+1} = A_t \tilde{s}_j + B_t \tilde{a}_j + c_t$ (ridge regression)

2 1-step energy bound (Lyapunov)

$\delta s_{t+1} \approx A_t \cdot \delta s_t \Rightarrow V(\delta s_{t+1}) \leqslant \sigma^2_{max}(A_t) \cdot V(\delta s_t)$

3 Normalize into sample weights

$\rho_t = \sigma_{max}(A_t)$, $r_t = \mathrm{clip}(\rho_t / \mathrm{median}\ \rho, 0.25, 4)$

Example curve (LIBERO-Object): risk spikes cluster in contact/manipulation and insertion phases (illustrative).

Note: only the state term A_t, no policy feedback $B_t \cdot J_\pi$; the authors state this is not a stability proof, only a policy-agnostic relative sensitivity measure.

Figure: Fig. 1; equations: Eq. (3)–(9)

Importance→plasticity: risk grads set per-tensor AdamW step

1

Per-sample action loss

$$\ell_i(\theta) = \text{mean}_{q \in Q_i} \text{CE}(\text{action token})$$

Action-token positions only; image, language, padding excluded

2

Risk-weighted tensor importance

$$\hat{I}_T = (1/N_k) \Sigma_i r_i^2 \cdot \| \nabla_T \ell_i \|^2_F / n_T$$

batch=1: square, then average; divide by n_T to remove size bias

3

Normalization + cumulative memory

$$I_T^{(k)} = \gamma \cdot I_T^{(k-1)} + N(\hat{I}_{new})_T \,, \quad \gamma = 1$$

N(·): divide by positive median to remove task-specific scale

4

Convert to learning rate

$$s_T = (1 + \beta \cdot \bar{I}_T)^{(-1/p)} \,, \quad \eta_T = \eta \cdot s_T$$

β sets suppression strength, p the curve shape

Key differences from EWC

No displacement penalty: no θ^* stored, no $\| \theta - \theta^* \|^2$ regularizer

Immediate protection: EWC penalty is 0 at a new task's start; DRAP shrinks steps from step one

Tunes optimizer steps: suppresses the actual AdamW step, not just raw gradient scale

Cost: important tensors may still drift via many small updates (Table I)

vs. EWC / CLASSP: DRAP adds a physical basis with the lightest state

Aspect	EWC	CLASSP	DRAP (ours)
History signal	Diagonal Fisher from old-task gradients	Accum. gradients/updates (powered)	Dynamics-risk-weighted per-sample gradients
Physical basis	None	None	**Local trajectory amplification risk**
Protection	Fisher-weighted quadratic displacement penalty	Update denominator + gradient threshold	**Per-tensor AdamW LR**
Ref. weights	Needs θ^*	None	None
When active	0 at new task start, grows with displacement	Immediately	Immediately
Granularity	Per-parameter	Usually per-parameter	Per-tensor (from per-sample grads)
Stored state	Diag. Fisher + reference params	Accum. adjustment stats	Accum. tensor importance only
Main risk	Inaccurate Fisher or over-regularization limits learning	Over-suppression blocks new tasks	Important tensors may still drift

Protocol: 1 ckpt, 3 LIBERO suites in sequence, zero replay

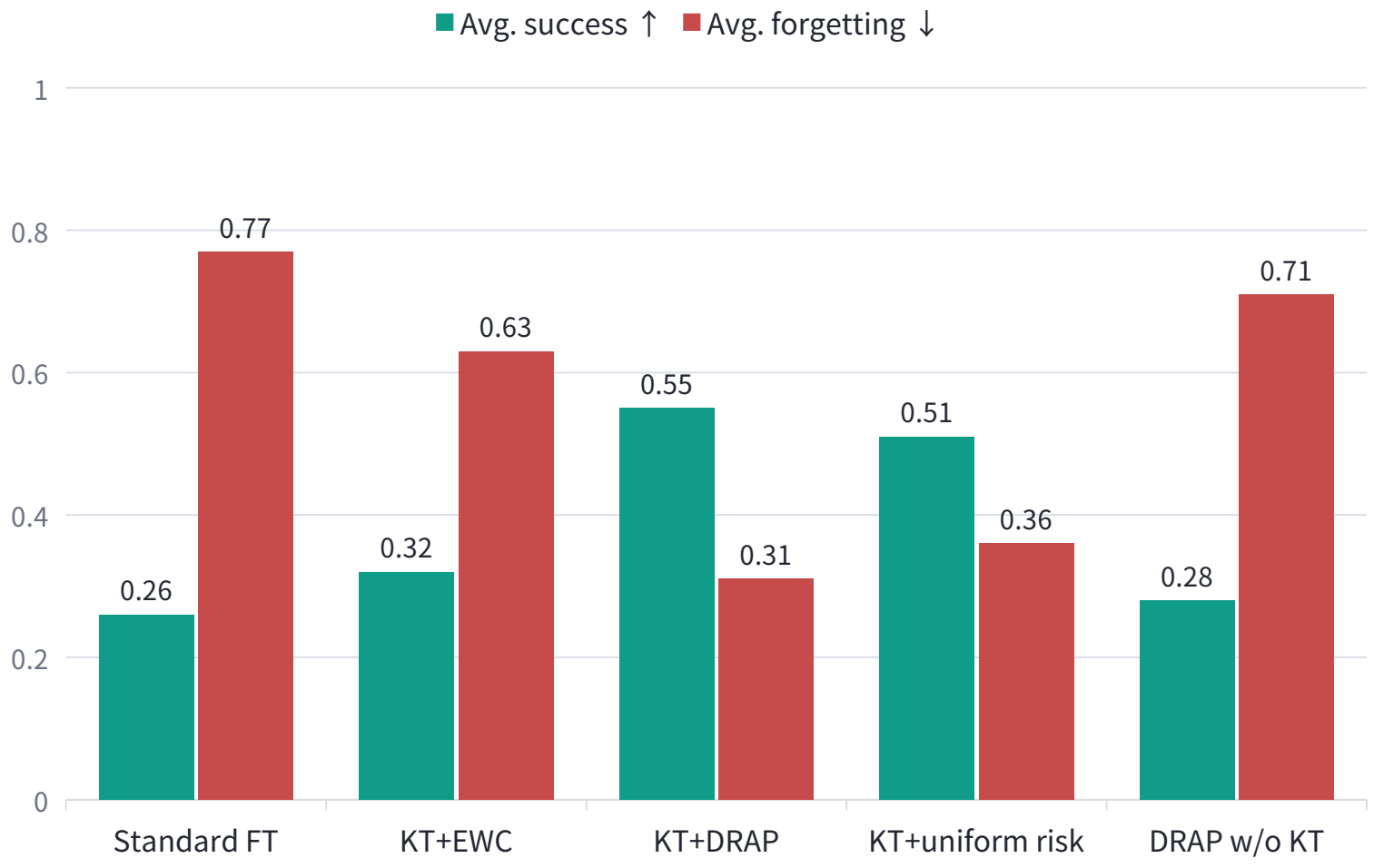
Setup

- Model: OpenVLA (7B, discrete action tokens); only the KappaTune-selected subnetwork is trained
- Each stage uses only current-suite data; no old demos stored
- After each stage, eval all seen suites → lower-triangular success matrix
- Pooled over 6 Gym evals × 50 episodes each
- Forgetting F = suite's best past success − current success (positive = forgetting)

5 compared configurations

- Standard sequential fine-tuning
- KappaTune + EWC
- **KappaTune + DRAP (full method)**
- KappaTune + uniform-risk DRAP ($r_i = 1$)
- DRAP without KappaTune

Results: after Goal, KT+DRAP top success, lowest forgetting



Note: KT = KappaTune

Key takeaways

0.55 vs 0.32

Avg. success: DRAP 0.23 higher than EWC

0.31 vs 0.63

Avg. forgetting: about half of EWC's

0.71

New-task Goal success matches standard FT; stability not bought with learning

Success matrix: FT keeps only newest suite; DRAP most old skills

Standard FT

After Object, Spatial 0.77→0.11; after Goal, old suites 0.00 / 0.07

EWC

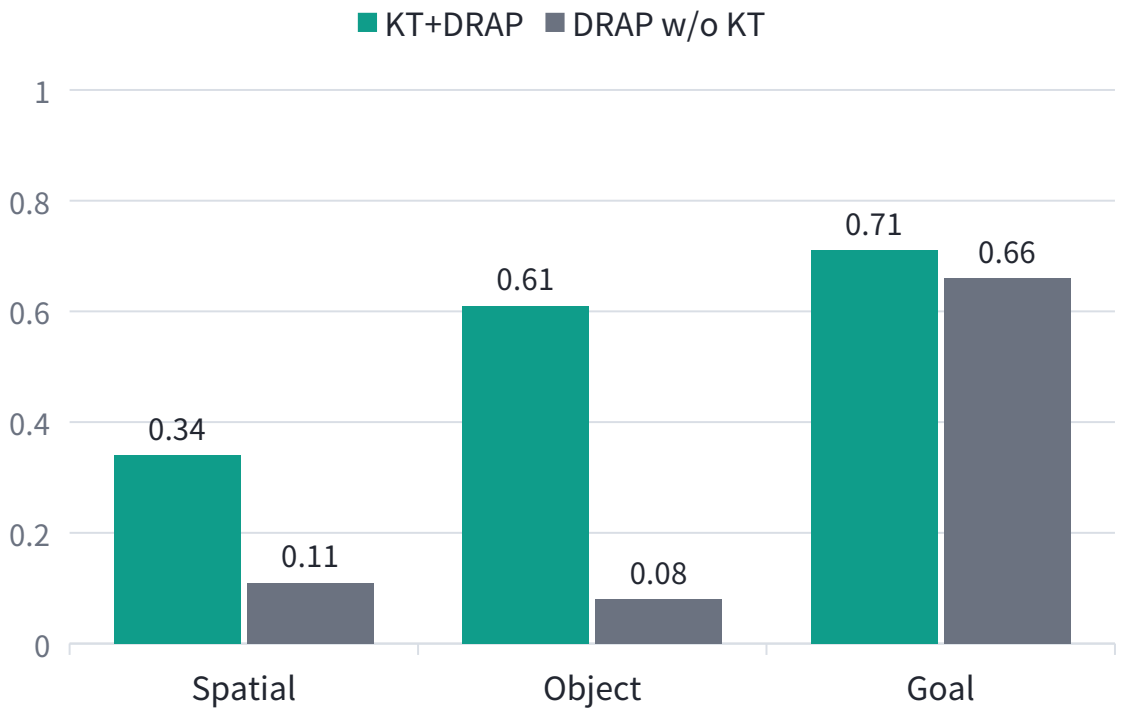
Spatial keeps 0.41 after the first switch, but only 0.03 after Goal

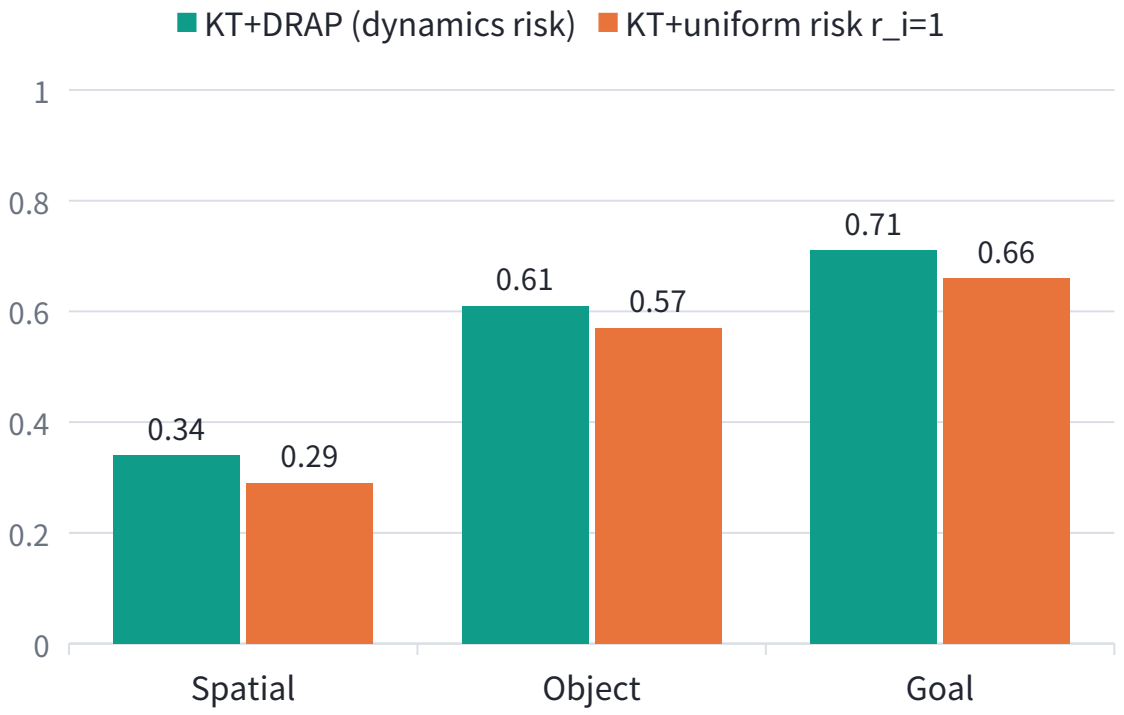
KT+DRAP

Spatial still 0.75 after Object; after Goal, old suites 0.34 / 0.61

Source: Table II (pooled over 6 × 50 episodes)

Ablation: KT stops late collapse; physical risk keeps a bit more





Without KappaTune: similar after Object (avg. success 0.77 for both), but after Goal old suites drop to 0.11 / 0.08, avg. forgetting 0.71, close to standard FT.

→ A larger trainable space means more interference paths; KappaTune and DRAP are complementary.

All risks set to 1: avg. forgetting 0.36 (full: 0.31), still far better than EWC's 0.63.

→ Most gains come from the importance→LR mechanism; physical risk adds ~0.04–0.05 retention, without hurting new-task learning.

Source: Tables II, III (after Goal)

Verdict: novel & lean; physical-risk gain needs firmer evidence

Strengths

- Brings physical consequences into CL parameter importance; clear intuition
- Lightweight: one scalar per tensor + AdamW param groups; no replay or reference weights
- Protection from a new task's first step, avoiding EWC's zero initial penalty
- Cross-suite (layout / object / goal semantics all change) with zero replay: a challenging setup
- Candid about risk-estimate limits: no policy feedback, one-step, locally linear

Limitations & Questions

- Physical vs. uniform risk differ by only ~0.03–0.05, with one task order and no multiple training seeds
- Absolute forgetting still sizable: Spatial 0.77 → 0.34
- Few baselines: no CLASSP, SI, MAS, LoRA-style methods, or small-replay upper bound
- Short sequence (3 suites); per-stage success ~0.7–0.8, below single-suite SOTA
- No quantitative check of r_t vs. actual failure locations; β, p and sensitivity not given in main text

Key Takeaways & Future Directions

1. CL in physical AI must weigh both model sensitivity and the physical consequences of behavioral errors.

2. DRAP: risk-weighted gradient importance → per-tensor AdamW LR; with KappaTune, avg. forgetting 0.77→0.31.

3. Most gains come from importance-driven plasticity control; dynamics risk adds extra retention.

Future directions proposed by the authors

- **Include control pathway**
  $B_t \cdot J_\pi$
- **Multi-step perturbation propagation**
- **Contact-aware / learned local dynamics**
- **Online estimation + CLASSP-style update thresholds**
- **Longer continual-learning sequences**